# EERoom

## A Browser-Based Electrical Engineering Simulation Platform

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## 1 Project Description

**EERoom** is a free web application where electrical engineering students run interactive simulations in the browser, with nothing to install and no account needed to start. It offers three tools, and every simulation can be saved to an account, reopened from a dashboard, and shared through a read-only link.

- **Ohm's Law Calculator:** solves V, I, or R from any two values and reports power.
- **Waveform Visualizer:** draws sine, square, sawtooth, and triangle signals and warns when the sample rate causes aliasing.
- **Logic Gate Sandbox:** wires seven gate types on a grid and builds the truth table and boolean expression live.

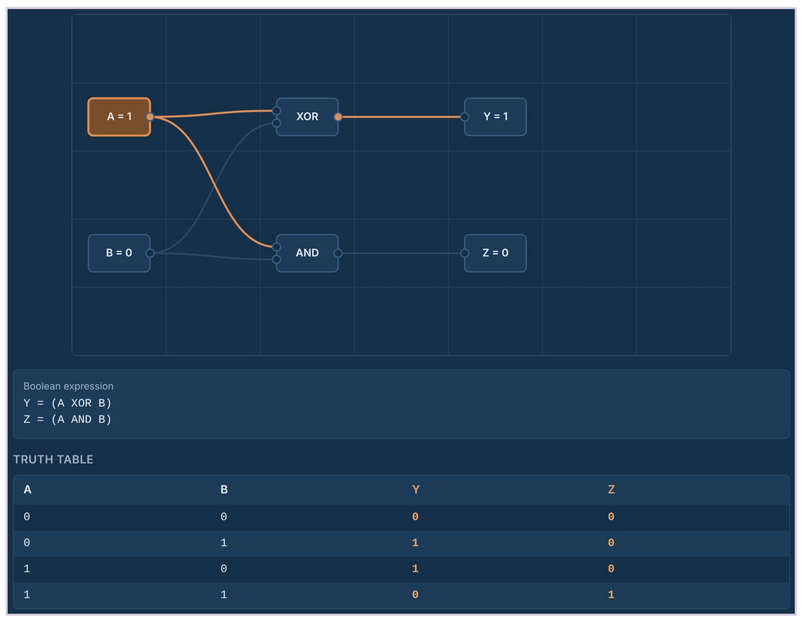



*Figure 1. A half adder built in the Logic Gate Sandbox. The truth table and expressions Y = A XOR B and Z = A AND B update as gates are wired.*

## 2 Project Justification

Circuit and signal concepts are abstract when they live only in equations. Students learn them faster when they can change a value and watch the result, but most simulation software is installed on lab computers, costs money, or takes a long time to learn.

EERoom gives students a free tool that opens in any browser, gives instant visual feedback, and lets them share work with an instructor through a link.

**1.5×** Students in traditional lecture courses were 1.5 times more likely to fail than students in active learning courses, across 225 STEM studies (Freeman et al., 2014).

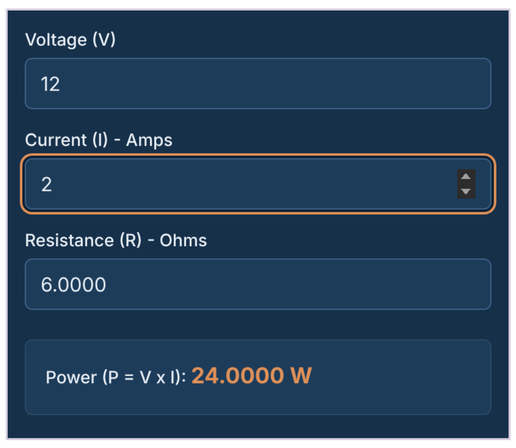



***Figure 2.*** *Ohm's Law Calculator. Entering 12 V and 2 A returns 6 Ω and 24 W instantly, with guards against division by zero.*

## 3 Requirements and Standards

### Key Functional Requirements

| Feature | The system shall… |
|---|---|
| **Accounts** | Register, log in, log out, and reset a forgotten password by email |
| **Guest use** | Use every tool without an account; keep the work after signing up |
| **Ohm's Law** | Calculate the third of V, I, R from any two, plus power |
| **Waveforms** | Plot 4 signal types with amplitude, frequency, phase, and offset |
| **Sampling** | Overlay the sampled signal and warn when it is aliased |
| **Logic gates** | Place 7 gate types, toggle inputs, and show the truth table |
| **Validation** | Flag unconnected pins and refuse feedback loops |
| **Projects** | Save, load, rename, and delete projects from a dashboard |
| **Sharing** | Create a read-only link that works without logging in |

### Product Specifications

| Layer | Technology |
|---|---|
| **Frontend** | React 18, Vite 5, Tailwind CSS 3, hosted on Vercel |
| **Backend** | Node.js and Express 4 REST API, hosted on Railway |
| **Database** | Supabase PostgreSQL |
| **Security** | Supabase Auth (bcrypt), JWT sessions, rate limiting |
| **Testing** | Vitest unit tests, Supertest API tests, and manual test cases |

### Standards Applied

- **IEEE 829:** test plans, cases, and results documentation
- **RFC 7519:** JSON Web Tokens for authenticated API calls
- **OWASP Top 10:** hashed passwords and auth rate limiting (10 tries per 15 min)
- **WCAG 2.2:** 4.5:1 text contrast and 24 px touch targets, tablet ready
- **SI units:** volts, amps, ohms, watts, hertz, and seconds
- **Nyquist-Shannon theorem:** sample rate must exceed twice the signal frequency

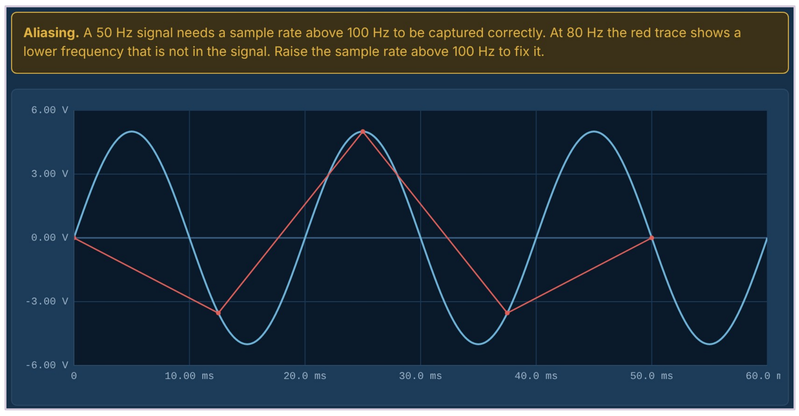



*Figure 3. A 50 Hz sine sampled at 80 Hz. The tool flags aliasing because 80 Hz is below the 100 Hz Nyquist rate.*

### Lessons Learned

- Cutting scope early let the remaining tools be finished, tested, and deployed.
- Keeping calculation logic in pure functions made it easy to unit test.
- API tests that send real requests check security rules a unit test cannot reach.
- A traceability matrix tied every requirement to a test case.
- Deploying early exposed CORS and environment setup issues before demo day.

## 4 Design Process

An Agile process with short iterations. Each milestone produced a document that guided the next one, and scope was reviewed after each code iteration.

1. **Proposal:** Problem, audience, and MVP scope
2. **Requirements:** 39 functional requirements and user stories
3. **Design:** Architecture, ER model, UI screens, security plan
4. **Iteration 1:** Accounts, project storage, Ohm's Law tool
5. **Iteration 2:** Waveform and logic tools, dashboard, sharing
6. **Scope review:** Requirements Revision 1.1 records every change
7. **Iteration 3:** Password reset, guest access, tablet layout
8. **Test and deploy:** 119 automated tests, live at eeroom.io

● CST-451 planning ● CST-452 build and test

### System Architecture

**Browser** (React client on Vercel) → HTTPS + JWT → **REST API** (Express server on Railway) → SQL → **Supabase** (PostgreSQL and Auth)

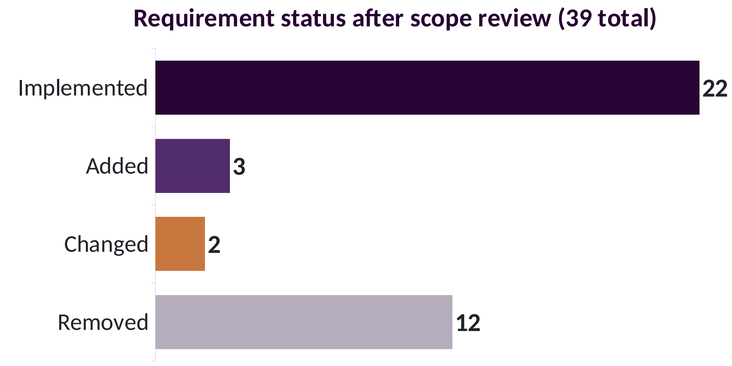
## 5 Data Summaries

**82/82** frontend unit tests passed

**37/37** API tests passed

**19/20** backlog items delivered

**99 h** actual vs. 113 h estimated




Removed items (circuit simulator, Google login, Stripe payments) were cut to finish and test the core tools well. 119 automated tests: 82 frontend unit tests and 37 API tests.

### Try It Live

**www.eeroom.io**

No account needed. Scan, open a tool, and start building.

## References

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